COMP1870

Matrices and Vectors

Theoretical Foundations Of Computer Science

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Acknowledgement

Following materials are referred for making this lecture note.

[1] Book *Elementary Linear Algebra* by William L. Perry.

[2] Book *Linear Algebra* by Jim Hefferon.

[3] Book *Linear Algebra* by Arak M. Mathai and Hans J. Haubold.

1 Matrices and Matrix algebra

1.1 Definition and Examples

Definition 1.1. A **matrix** is a rectangular array of numbers (or other mathematical objects), called the **entries** or **elements** of the matrix. A matrix with m rows and n columns is called an $m \times n$ matrix or m-by-n matrix. We also say that A is of **order** $m \times n$.

In this module we consider the case where all the entries in the matrices are real numbers.

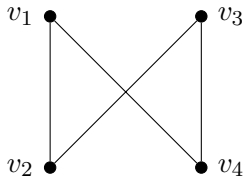
Example 1.2. Two matrices A and B are given below. A is a 4×4 matrix and B is a 3×4 matrix.

$$A = \begin{bmatrix} 10 & 1 & 0 & 9 \\ 12.4 & 6 & 1 & 0 \\ 0 & 6 & 0 & 8 \\ 1 & 3.14 & 1 & 0 \end{bmatrix} \qquad B = \begin{bmatrix} 0 & 6 & 3 & 1 \\ 1 & 4 & 1 & 0 \\ 7 & 0 & 10 & 20 \end{bmatrix}$$

For an $m \times n$ matrix A and positive integers $i \in \{1, \ldots, m\}$ and $j \in \{1, \ldots, n\}$, we use $A[i, j]$ to denote the entry in the ith row and jth column of A. For example, $A[2, 3] = 1$ in the matrix A given in Example 1.2.

Below are some examples, where matrices are used.

Example 1.3. Adjacency matrix of a graph, where rows and columns corresponding to the vecrtices of the graph.



$$\begin{bmatrix} 0 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 \end{bmatrix}$$

Example 1.4. The assignment problem : Imagine the following situation. We have access to three servers in the cloud, and three tasks to be completed by those servers. Each server is going to process exactly one task. ('Exactly one' means 'one, and no more than one'.) But the servers are not identical. The difficulty and the cost of completing a task may vary from server to server. We summarise the situation with the array below, which

shows the cost of completing each task.

	Task 1	Task 2	Task 3
Server 1	110	20	45
Server 2	30	80	125
Server 3	40	65	95

Now our job is to assign each of the three tasks to one of the three servers. We would like to do so in such a way that the total cost is as small as possible.

We can represent the input to the assignment problem using a matrix where the rows corresponding to the servers, the columns corresponding to the tasks, and each entry represents the corresponding cost of performing the task in the server.

$$\begin{bmatrix} 110 & 20 & 45 \\ 30 & 80 & 125 \\ 40 & 65 & 95 \end{bmatrix}$$

Example 1.5. In image processing, an image can be represented as a matrix of pixels, where each pixel corresponds to an element in the matrix.

Definition 1.6. Two matrices A and B are said to be equal, written $A = B$, if they have the same order and their corresponding entries are equal. That is, $A[i, j] = B[i, j]$ for all i and j.

Example 1.7. Two matrices A and B given below are equal.

$$A = \begin{bmatrix} 0 & 12/2 & 9/3 & 1.0 \\ 1 & 8/2 & 1 & 0 \\ 7 & 0 & 10 & 20 \end{bmatrix} \qquad B = \begin{bmatrix} 0 & 6 & 3 & 1 \\ 1 & 4 & 1 & 0 \\ 7 & 0 & 20/2 & 20 \end{bmatrix}$$

Definition 1.8. A matrix with same number of rows and columns is called a square matrix. For an $n \times n$ square matrix A, the entries $A[1, 1], A[2, 2], \ldots, A[n, n]$ forms the **main diagonal** of A.

Matrix A in Example 1.2 is a square matrix.

Definition 1.9. If all the elements in a matrix are zeros, then it is called a **null matrix** and it is denoted by $\mathbf{O}$.

Definition 1.10. An $n \times n$ matrix A is a **diagonal matrix**, if $A[i, j] = 0$ for all $i \neq j$ and at least one $A[i, i] \neq 0$, $i = 1, \ldots, n$.

Example 1.11. The below matrix is an example of a diagonal matrix.

$$\begin{bmatrix} 10 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 7 & 0 \\ 0 & 0 & 0 & 3 \end{bmatrix}$$

A convenient notation for a diagonal matrix is

$$D = \mathsf{diag}(d_1, \ldots, d_n)$$

which means D is a $n \times n$ diagonal matrix with the diagonal entries $d_1, \ldots, d_n$.

Definition 1.12 (Identity matrix)**.** A diagonal matrix where all the diagonal elements equal to 1 is called an identity matrix. An $n \times n$ identity matrix is denoted by I_n.

Example 1.13. A 4×4 identity matrix I_4 is given bleow.

$$I_4 = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Definition 1.14 (Transpose of a matrix)**.** If the ith row of an $m \times n$ matrix A is written as the ith column for all $i \in \{1, \ldots, m\}$, then the new matrix obtained is called the transpose of A and it is denoted by A^T.

Example 1.15. Below is an example of a matrix and its transpose.

$$A = \begin{bmatrix} 0 & 1 & 7 \\ 6 & 4 & 0 \\ 3 & 1 & 10 \\ 1 & 0 & 20 \end{bmatrix} \qquad A^T = \begin{bmatrix} 0 & 6 & 3 & 1 \\ 1 & 4 & 1 & 0 \\ 7 & 0 & 10 & 20 \end{bmatrix}$$

Definition 1.16 (Symmetric matrix)**.** An $n \times n$ square matrix A is a symmetric matrix if $A[i, j] = A[j, i]$ for all $i, j \in \{1, 2, \ldots, n\}$. That is $A^T = A$.

The adjacency matrix of a graph is a symmetric matrix.

Definition 1.17 (Skew symmetric matrix)**.** An $n \times n$ square matrix A is a skew symmetric matrix if $A[i, j] = -A[j, i]$ for all $i, j \in \{1, 2, \ldots, n\}$. That is $A^T = -A$.

Notice that in a skew symmetric matrix all the diagonal entries are zeros.

1.2 Basic operations

Definition 1.18 (Scalar multiplication)**.** A **scalar multiple** cA of a matrix A, where c is a real number called the **scalar**, is a matrix obtained from A where each entry is multipled by c.

$$c\begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{n2} & \cdots & a_{mn} \end{bmatrix} = \begin{bmatrix} ca_{11} & ca_{12} & \cdots & ca_{1n} \\ ca_{21} & ca_{22} & \cdots & ca_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ ca_{m1} & ca_{n2} & \cdots & ca_{mn} \end{bmatrix}$$

Definition 1.19 (Addition of matrices)**.** If A and B are both $m \times n$ matrices, then they are **conformable for addition** and the sum is defined by $(A + B)[i, j] = A[i, j] + B[i, j]$ for all i and j.

Example 1.20. Example of matrix addition.

$$\begin{bmatrix} 10 & 2 & 4 \\ 30 & 8 & 12 \end{bmatrix} + \begin{bmatrix} 1 & 4 & 3 \\ -3 & -1 & 5 \end{bmatrix} = \begin{bmatrix} 11 & 6 & 7 \\ 27 & 7 & 17 \end{bmatrix}$$

Theorem 1.21. *Let A, B, C be three matrices of order $m \times n$, and let α and β be two scalars.*

- $-A = -1 \cdot A$.
- $A - A = \mathbf{O}$.
- $A + \mathbf{O} = A$.
- $A + B = B + A$. *That is, matrix addition is commutative.*
- $A + (B + C) = (A + B) + C$. *That is, matrix addition is associative.*
- $\alpha(A + B) = \alpha A + \alpha B$. *That is, scalar multiplication distributes over addition.*
- $(\alpha + \beta)A = \alpha A + \beta A$. *That is, scalar multiplication distributes over addition.*

The proof of each statement in the above theorem reduces to properties of real numbers by looking at each entry in the matrices in the either side of the equality. This is left as an exercise.

Definition 1.22 (Matrix multiplication)**.** For an $m \times n$ matrix A and an $n \times p$ matrix B,

$$A = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix}, \quad B = \begin{bmatrix} b_{11} & b_{12} & \cdots & b_{1p} \\ b_{21} & b_{22} & \cdots & b_{2p} \\ \vdots & \vdots & \ddots & \vdots \\ b_{n1} & b_{n2} & \cdots & b_{np} \end{bmatrix}$$

the **matrix product** $C = A \cdot B$ (also denoted as $C = AB$) is defined to be the $m \times p$ matrix,

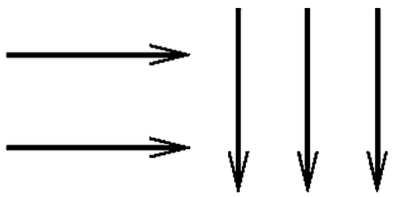
$$C = \begin{bmatrix} c_{11} & c_{12} & \cdots & c_{1p} \\ c_{21} & c_{22} & \cdots & c_{2p} \\ \vdots & \vdots & \ddots & \vdots \\ c_{m1} & c_{m2} & \cdots & c_{mp} \end{bmatrix}$$

such that

$$c_{ij} = a_{i1}b_{1j} + a_{i2}b_{2j} + \cdots + a_{in}b_{nj} = \sum_{k=1}^{n} a_{ik}b_{kj},$$

for every $i \in \{1, \ldots, m\}$ and $j \in \{1, \ldots, p\}$.

Symbolically matrix multiplication can be expressed as



Example 1.23. Let A and B be the matrices given below.

$$A = \begin{bmatrix} 3 & 1 \\ 1 & 2 \end{bmatrix} \qquad B = \begin{bmatrix} 1 & 0 \\ 3 & 8 \end{bmatrix}$$

Then,

$$AB = \begin{bmatrix} 3 & 1 \\ 1 & 2 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 3 & 8 \end{bmatrix} = \begin{bmatrix} 3 \cdot 1 + 1 \cdot 3 & 3 \cdot 0 + 1 \cdot 8 \\ 1 \cdot 1 + 2 \cdot 3 & 1 \cdot 0 + 2 \cdot 8 \end{bmatrix} = \begin{bmatrix} 6 & 8 \\ 7 & 16 \end{bmatrix}$$

$$BA = \begin{bmatrix} 1 & 0 \\ 3 & 8 \end{bmatrix} \begin{bmatrix} 3 & 1 \\ 1 & 2 \end{bmatrix} = \begin{bmatrix} 1 \cdot 3 + 0 \cdot 1 & 1 \cdot 1 + 0 \cdot 2 \\ 3 \cdot 3 + 8 \cdot 1 & 3 \cdot 1 + 8 \cdot 2 \end{bmatrix} = \begin{bmatrix} 3 & 1 \\ 17 & 19 \end{bmatrix}$$

From the above example, we deduce that for two matrices A and B, although both AB and BA are defined, AB may not be equal to BA.

Example 1.24. Suppose you have the following system of linear equations.

$$\begin{aligned} 5x_1 + 3x_3 + 12x_4 &= 9 \\ 2x_1 + 15x_2 + 14x_3 + x_4 &= 100 \\ 7x_3 + 2x_4 &= 21 \\ x_1 + 2x_2 &= 8 \end{aligned}$$

Then, you can write it in matrix format as follows.

$$\begin{bmatrix} 5 & 0 & 3 & 12 \\ 2 & 15 & 14 & 1 \\ 0 & 0 & 7 & 2 \\ 1 & 2 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} 9 \\ 100 \\ 21 \\ 8 \end{bmatrix}$$

Theorem 1.25. *Let A, B and C be three matrices. For the statement below, assume that they are in correct order for which addition and multiplication are defined.*

- *$A(B+C)=AB+AC$.*
- *$(A+B)C=AC+BC$.*
- *$(AB)C=A(BC)$.*

Exercise 1.26. Prove Theorem 1.25.

Theorem 1.27. *Let A be an $n \times n$ matrix. Then,*

- *$AI_n = I_nA = A$, and*
- *$I_nI_n = I_n$.*

Exercise 1.28. Prove Theorem 1.27.

Theorem 1.29. *Let A be an $m \times n$ matrix and B be an $n \times p$ matrix. Then, prove that $(AB)^T = B^TA^T$.*

Proof. Let $C = AB$. The order of C is $m \times p$ and the order of $(AB)^T$ (which is equal to C^T) is $p \times m$. Notice that B^T is an $p \times n$ matrix and A^T is a $n \times m$ matrix. Let $D = B^TA^T$. The order of D is $p \times m$. We want to prove that $C^T = D$.

Let $i \in \{1, \ldots, p\}$ and $j \in \{1, \ldots, m\}$ be two fixed integers. Then $C^T[i, j] = C[j, i]$. By the definition of matrix multiplication,

$$C^T[i, j] = C[j, i] = \sum_{k=1}^{n} A[j, k]B[k, i] \tag{1.1}$$

By the definition of matrix multiplication,

$$\begin{aligned} D[i, j] &= \sum_{k=1}^{n} B^T[i, k]A^T[k, j] \\ &= \sum_{k=1}^{n} B[k, i]A[j, k] \\ &= \sum_{k=1}^{n} A[j, k]B[k, i] \end{aligned} \tag{1.2}$$

The last equality follows from the fact that multiplication of real numbers is commutative. From (1.1) and (1.2), we get that $C^T = D$ and thus $(AB)^T = B^T A^T$. □

Algorithms for matrix multiplication

MATRIX MULTIPLICATION
Input: An $m \times n$ integer matrix A and an $n \times p$ integer matrix B.
Output: AB

Algorithm 1 Algorithm for MATRIX MULTIPLICATION

Input: An $m \times n$ matrix A and an $n \times p$ matrix B.
1: Let C be an $m \times p$ matrix with all entries equal to zero.
2: **for** $i = 1$ to m **do**
3: **for** $j = 1$ to p **do**
4: **for** $k = 1$ to n **do**
5: $C[i, j] := C[i, j] + (A[i, k] \cdot A[k, j])$
6: **end for**
7: **end for**
8: **end for**
9: **Output** C

The above algorithm computes the matrix product $C = AB$. There are $m \cdot p$ entries in C. To compute each entry, we do n integer multiplication and n integer addition. Thus the total number of integer addition and multiplication operations is $m \cdot p \cdot (2n)$. When, $n = m = p$, the total number of integer addition and multiplication operations is $2n^3$.

In 1969, V Strassen provided an algorithm with $cn^{\log_2 7} = cn^{2.8074}$ operations, where c is a constant independent of n. After that, there has been many research to improve the exponent of n. The current best bound $cn^{2.371552}$ is obtained in 2024, by V V Williams, Y Xu, Z Xu, and R Zhou. It is a long standing open problem to improve the exponent to 2.

Matrix Chain Multiplication. Suppose we are given three integer matrices A, B and C where A is a 10×30 matrix, B is a 30×5 matrix, and C is a 5×60 matrix. We want to compute ABC. We have two options. We may first multiply A and B, then the multiply the resultant matrix with C. The second option is, first multiply B and C and then left multiply the resultant matrix with A. To compute $(AB)C$ we need $2(10 \times 30 \times 5) + 2(10 \times 5 \times 60) = 3000 + 6000 = 9000$ integer addition and multiplication operations. On the other hand, for computing $A(BC)$ we need $2(30 \times 5 \times 60) + 2(10 \times 30 \times 60) = 18000 + 36000 = 54000$ integer addition and multiplication operations. In the matrix chain multiplication problem, we are a given sequence of matrices and the objective is to find a sequence of matrix multiplication involved so that we use minimum number of operations. The above example mentions that the number of operations vary drastically based the choices we make. Here, we do not discuss about how we can compute the sequence of matrix multiplication operations. But, we want to mention that it is important to design algorithms that are efficient in terms of the number of operations.

2 Vectors

2.1 Definition and Basic operations

In this section first we give a simple definition for vectors and then introduce geometric interpretation of it in the next subsection.

Definition 2.1. A **column vector**, often just called a **vector**, is a matrix with a single column. A matrix with a single row is a **row vector**. The entries of a vector are called **components**. A column or row vector whose components are all zeros is a **zero vector**.

Example 2.2. See examples of vector, row vector, and zero vector below.

$$\begin{bmatrix} 0 & 1 & 7 \end{bmatrix} \qquad \begin{bmatrix} 0 \\ 1 \\ 3.1 \\ 7 \end{bmatrix} \qquad \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix} \qquad \begin{bmatrix} 0 & 0 & 0 \end{bmatrix}$$

row vector vector zero vector zero vector

We use lower case letters overlined with an arrow to denote a vector. For example,

$$\vec{v} = \begin{bmatrix} 0 \\ 1 \\ 3.1 \\ 7 \end{bmatrix}$$

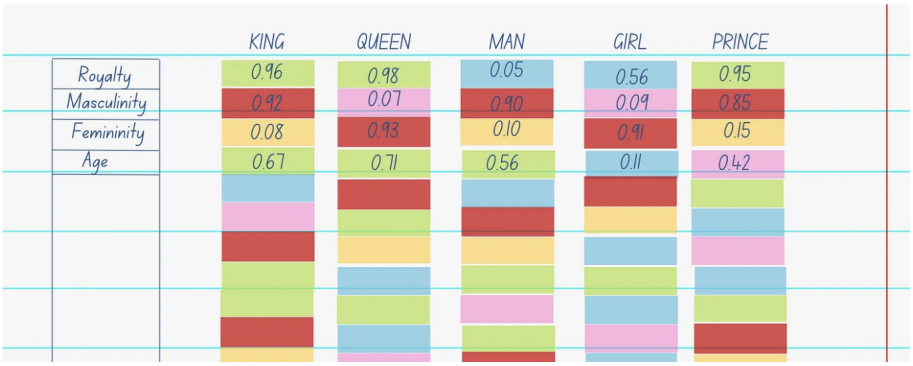
A zero vector is denoted by $\vec{0}$.

Example 2.3. Following are some examples of vectors in computer science.

- In machine learning and programming it is often convenient to use vectors. For instance, in the housing price prediction algorithms, we might have features like year of construction of the house, number of bedrooms, area, size of backyard, and size of garage. This would translates to vectors such as

$$\begin{bmatrix} 2000 & 2 & 65 & 40 & 9 \end{bmatrix}, \begin{bmatrix} 2009 & 3 & 95 & 50 & 8 \end{bmatrix}.$$

- Word2vec is a technique in natural language processing (NLP) for obtaining vector representations of words. These vectors capture information about the meaning of the word based on the surrounding words. The word2vec algorithm estimates these representations by modeling text in a large corpus. Word2Vec creates a representation of each word into a vector. Words used in similar contexts or having semantic relationships are captured effectively through their "closeness in the vector space"- i.e., similar words will have similar word vectors. Word2vec was created, patented, and published in 2013 by a team of researchers led by Tomas Mikolov at Google. A hypothetical example is given below. [Reference: Blog by Manan Suri]



- In robotics, vectors can represent position, velocity, force, etc.

Definition 2.4. The **scalar multiplication** of a real number r and the vector $\vec{v}$ is defined as follows.

$$r \cdot \vec{v} = r \begin{bmatrix} v_1 \\ v_2 \\ \vdots \\ v_n \end{bmatrix} = \begin{bmatrix} rv_1 \\ rv_2 \\ \vdots \\ rv_n \end{bmatrix}$$

Example 2.5.

$$3 \cdot \begin{bmatrix} 4 \\ 2 \\ -1 \\ 7 \end{bmatrix} = \begin{bmatrix} 3 \cdot 4 \\ 3 \cdot 2 \\ 3 \cdot -1 \\ 3 \cdot 7 \end{bmatrix} = \begin{bmatrix} 12 \\ 6 \\ -3 \\ 21 \end{bmatrix}$$

Definition 2.6. The **vector sum** of $\vec{u}$ and $\vec{v}$ is the vector of the sums.

$$\vec{v} + \vec{u} = \begin{bmatrix} v_1 \\ v_2 \\ \vdots \\ v_n \end{bmatrix} + \begin{bmatrix} u_1 \\ u_2 \\ \vdots \\ u_n \end{bmatrix} = \begin{bmatrix} v_1 + u_1 \\ v_2 + u_2 \\ \vdots \\ v_n + u_n \end{bmatrix}$$

Vector sum of $\vec{u}$ and $\vec{v}$ is defined only if the number of components in it are equal.

Example 2.7.

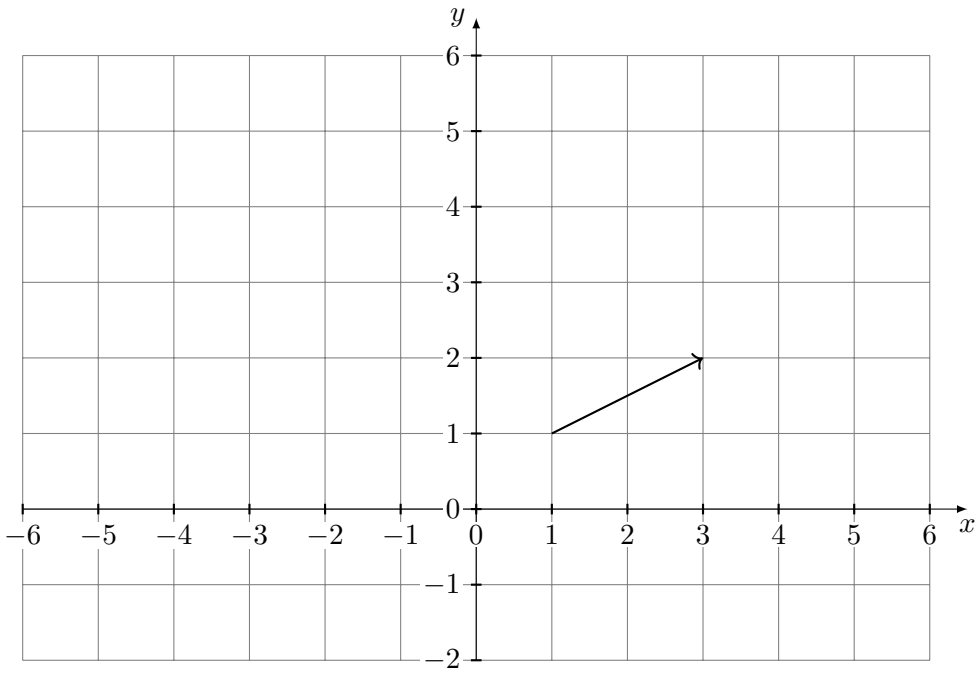
$$\begin{bmatrix} 4 \\ 2 \\ -1 \\ 7 \end{bmatrix} + \begin{bmatrix} 3 \\ -8 \\ 6 \\ 5 \end{bmatrix} = \begin{bmatrix} 4+3 \\ 2-8 \\ -1+6 \\ 7+5 \end{bmatrix} = \begin{bmatrix} 7 \\ -6 \\ 5 \\ 12 \end{bmatrix}$$

2.2 Geometric Interpretation

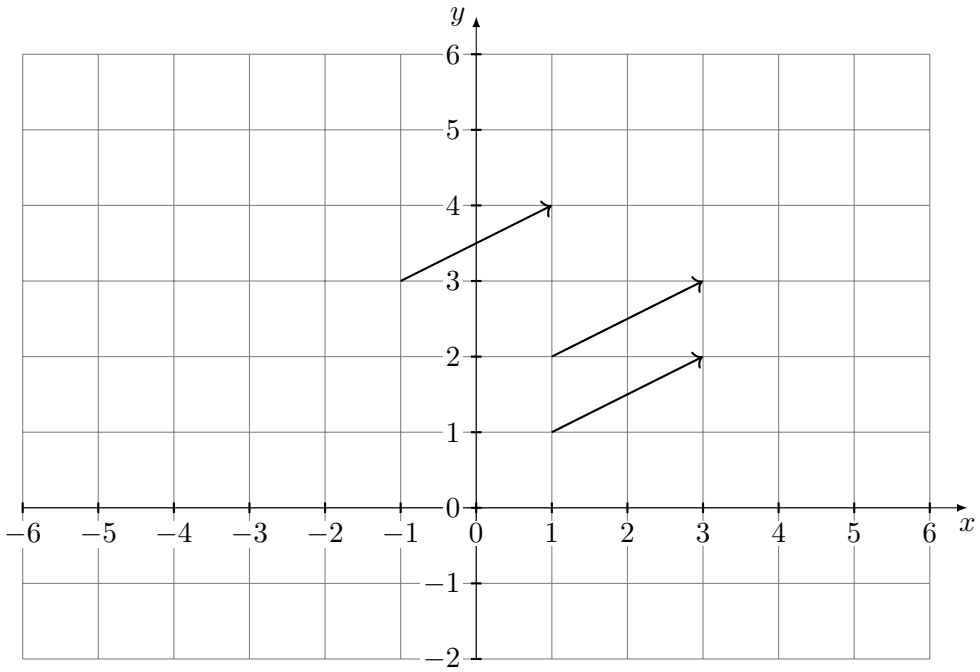
A **Euclidean plane** is a geometric space where each point is represented by a pair of ordered real numbers. We use $\mathbb{R}^2$ to denote the Euclidean plane.

Definition 2.8. An object in $\mathbb{R}^2$ that is comprised of **magnitude** (or **length**) and **direction** is called a **vector** in $\mathbb{R}^2$.

We can draw a vector as having some length and pointing in some direction.



But, then the following three vectors have the equal length and same direction.



In fact, these vectors are **equal** because they have equal lengths and equal directions. Again, those vectors are not just alike, they are equal. How can things that are in different places be equal? Think of a vector as representing a displacement. One of these vector starts at the point $(1, 1)$ and has a displacement of 2 in the direction of x-axis and 1 in the direction of y-axis. Another one starts at $(1, 2)$ and has a displacement of 2 in the direction of x-axis and 1 in the direction of y-axis. This is the case with the vector starting at $(-1, 3)$. So, irrespective of their starting position, the displacement in both the direction is equal.

In other words, there are three vectors in the above figure: (i) extending from $(1, 1)$ to $(3, 2)$, (ii) extending from $(1, 2)$ to $(3, 3)$ and (iii) extending from $(-1, 3)$ to $(1, 4)$. Now for the first vector, the change in the first component and the second component is $(3 - 1, 3 - 2) = (2, 1)$. In fact, for other two vectors also, the change in the first component and the second component is $(2, 1)$. Thus, we say that these vectors are equal.

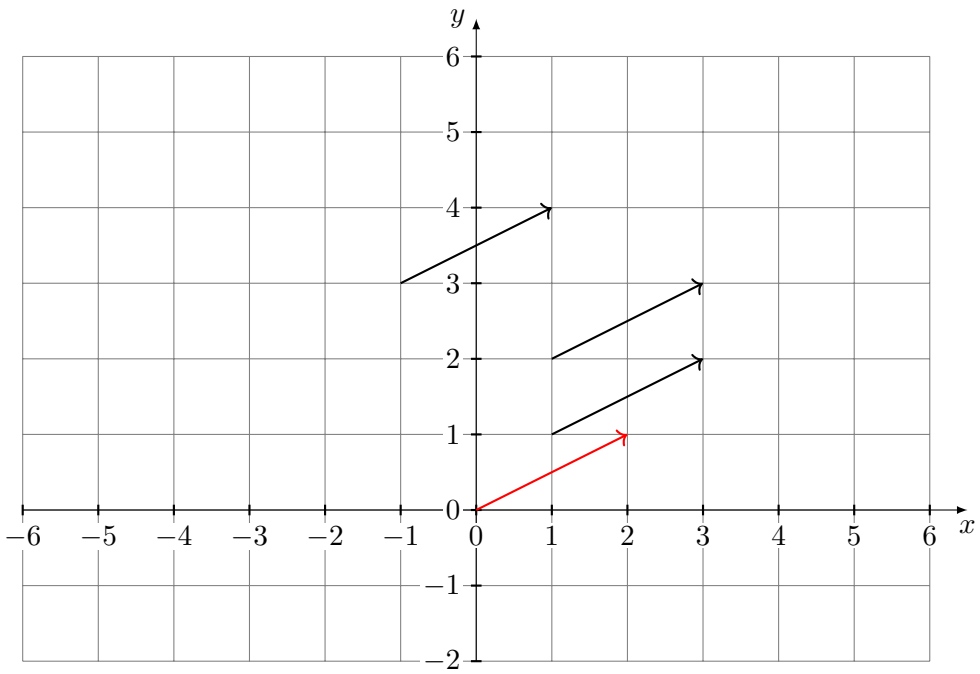
More generally, vectors in the plane are the same if and only if they have the same change in first components and the same change in second components: the vector extending from (a_1, a_2) to (b_1, b_2) equals the vector from (c_1, c_2) to (d_1, d_2) if and only if $b_1 - a_1 = d_1 - c_1$. and $b_2 - a_2 = d_2 - c_2$.

Instead of saying that "the vector start at (a_1, a_2), and extend to

(b_1, b_2)", we may describe the vector as

$$\begin{bmatrix} b_1 - a_1 \\ b_2 - a_2 \end{bmatrix}$$

In other words, we may assume the satrting point of a vector is origin $(0, 0)$. We draw the arrow as starting at the origin, and then we say it is in the **canonical position** (or **natural position** or **standard position**). For example, the vector drawn in red in the below figure is in canonical position. The vector which is not in canonical position is called **free vector**.



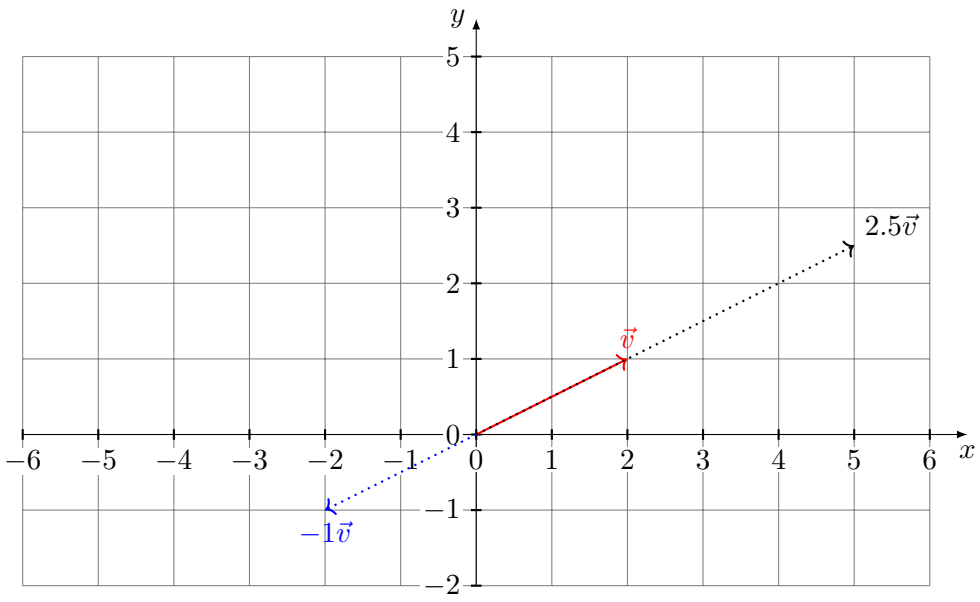
When $\vec{v} = \begin{bmatrix} v_1 \\ v_2 \end{bmatrix}$ is in canonical position, then $\vec{v}$ extend from origin $(0, 0)$ to (v_1, v_2). Thus, we may use $\mathbb{R}^2$ to denote the set of points in the plane $\{(a_1, a_2) \mid a_1, a_2 \in \mathbb{R}\}$ as well as the set of vectors in the plane $\{\begin{bmatrix} a_1 \\ a_2 \end{bmatrix} \mid a_1, a_2 \in \mathbb{R}\}$.

Previously, we defined vectors and vector operations like scalar multiplication and vector sum in an algebraic sense.

Definition 2.9. The **scalar multiplication** of a real number r and a vector $\vec{v}$ in plane is defined as follows.

$$r \cdot \vec{v} = \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} rv_1 \\ rv_2 \end{bmatrix}$$

We can understand the scalar multiplication geometrically now. That is, since the vector $\vec{v}$ represents a displacement, $2.5\vec{v}$ represents a displacement in the same direction, but 2.5 times far. Also, $-1\vec{v}$ represents the displacement of same length but in the opposite direction.

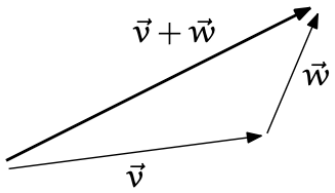


Lemma 2.10. *For a vector $\vec{v}$ and two scalars r_1 and r_2,*

$$(r_1 + r_2)\vec{v} = r_1\vec{v} + r_2\vec{v}.$$

Exercise 2.11. Prove Lemma 2.10.

We can understand vector sum geometrically now. Since $\vec{v}$ and $\vec{w}$ represent displacements, $\vec{v} + \vec{w}$ represents those displacements combined. This is depicted in the below figure.

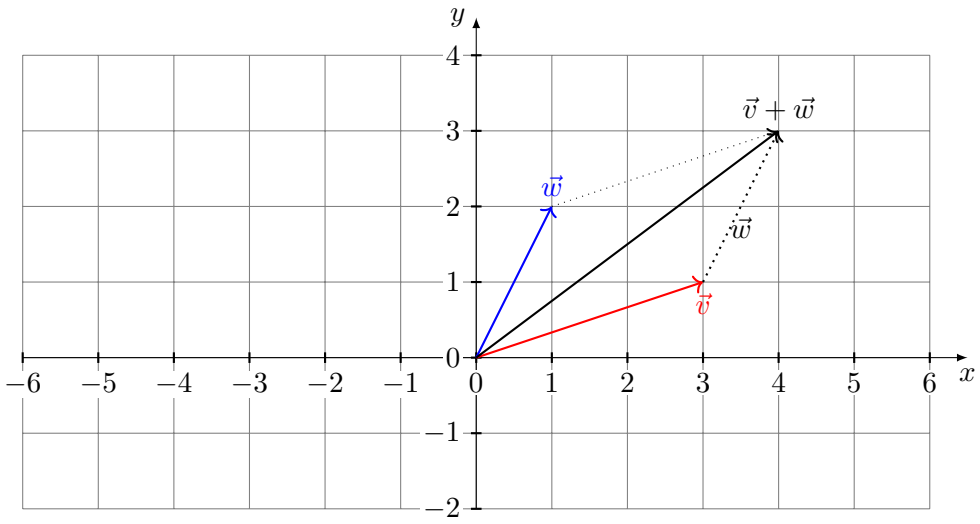


Notice that in the above figure, both $\vec{v}$ and $\vec{w}$ together cannot be in canonical position. Because, if the starting point of $\vec{v}$ is origin, then the starting point of $\vec{w}$ is not origin. But, how does this interpretation translate to the algebraic definition (given below) of the vector sum.

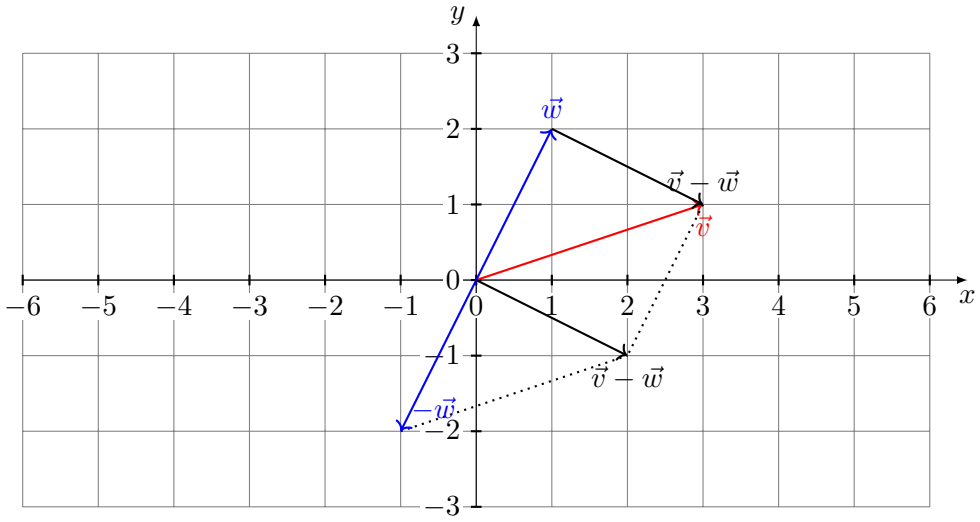
Definition 2.12. The **vector sum** of $\vec{v}$ and $\vec{w}$ in plane is defined as follows.

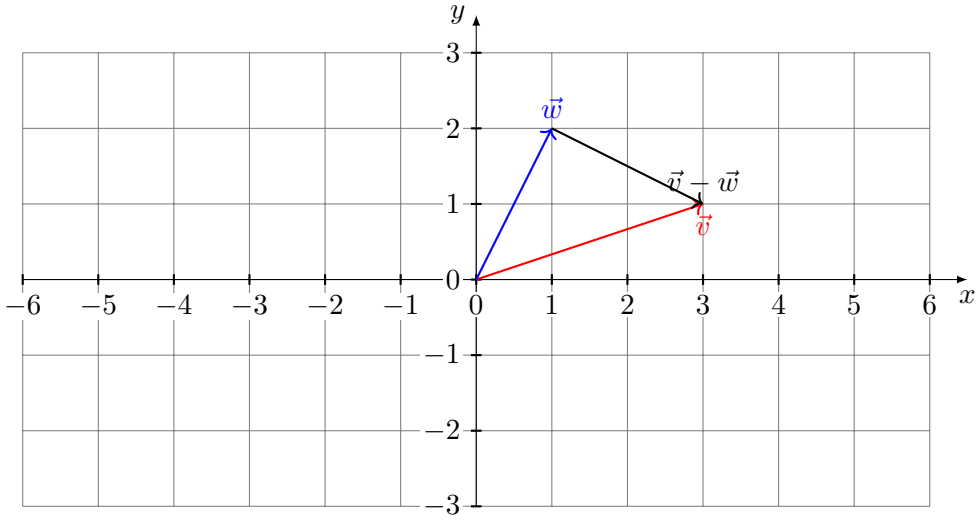
$$\vec{v}+\vec{w} = \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} + \begin{bmatrix} w_1 \\ w_2 \end{bmatrix} = \begin{bmatrix} v_1 + w_1 \\ v_2 + w_2 \end{bmatrix}$$

To observe that the geometric interpretation is same as the algebraic definition, we need to understand the vector sum with **parallelogram rule**. That is, first draw $\vec{v}$ and $\vec{w}$ in canonical position. Then, extend it to a parallelogram, where one "line" is equal to the vector $\vec{w}$. Notice that the vector sum $\vec{v}+\vec{w}$ extends along the diagonal to the far corner. Moreover, observe that the first component of $\vec{v}+\vec{w}$ is the sum of the first component of $\vec{v}$ and the first component of $\vec{w}$. The case for the second component is similar.



Now let us discuss geometric representation of the vector $\vec{v}-\vec{w}$. See the figure below. The vector $\vec{v}-\vec{w}$ is same as $\vec{v}+(-\vec{w})$. Then, we get the vector drawn with black color in the canonical form as $\vec{v}-\vec{w}$. You can see a free vector drawn in black color with the same length and direction as $\vec{v}-\vec{w}$ in the the canonical form. Thus, another interpretation of $\vec{v}-\vec{w}$ as a side of the triangle as shown in the next figure.





Our discussion here can be extended to $\mathbb{R}^3$. In fact it can be extended to

even higher space $\mathbb{R}^n$. A free vector that starts at $(a_1, \ldots, a_n)$ and extends to $(b_1, \ldots, b_n)$ is represented by the column

$$\begin{bmatrix} b_1 - a_1 \\ b_2 - a_2 \\ \vdots \\ b_n - a_n \end{bmatrix}$$

We mentioned that a vector has a magnitude or length which is the length of the arrow we draw.

Definition 2.13. The **length** of a vector $\vec{v} = [v_1, \ldots, v_n]^T \in \mathbb{R}^n$, denoted by $|\vec{v}|$ is the square root of the sum of the squares of its components. That is,

$$|\vec{v}| = \sqrt{v_1^2 + v_2^2 + \ldots + v_n^2}.$$

The geometric interpretation of the length of a vector in $\mathbb{R}^2$ follows from Pythagorean Theorem: for a point (a_1, a_2) in $\mathbb{R}^2$, the straight line distance from $(0, 0)$ to (a_1, a_2) is $\sqrt{a_1^2 + a_2^2}$.

Definition 2.14. The **dot product** (or **inner product** or **scalar product**) of two vectors $\vec{u} = [u_1, \ldots, u_n]$ and $\vec{v} = [v_1, \ldots, v_n]$, denoted by $\vec{u} \cdot \vec{v}$, is

$$\sum_{i=1}^{n} u_i v_i.$$

It is important to note that dot product of two vectors is a real number and not a vector. Notice that the dot product of $\vec{u} = [u_1, \ldots, u_n]$ and $\vec{v} = [v_1, \ldots, v_n]$ is same as the matrix product

$$[u_1, \ldots, u_n] \begin{bmatrix} v_1 \\ v_2 \\ \vdots \\ v_n \end{bmatrix}$$

Theorem 2.15. *Let $\vec{u}, \vec{v}, \vec{w}$ are vectors in $\mathbb{R}^n$, and $\alpha, \beta \in \mathbb{R}$ are two scalars. Then the following equations are true.*

- $\vec{u} \cdot \vec{v} = \vec{v} \cdot \vec{u}$
- $\vec{u} \cdot (\vec{v} + \vec{w}) = \vec{u} \cdot \vec{v} + \vec{u} \cdot \vec{w}$

- $(\alpha\vec{u}) \cdot (\beta\vec{v}) = \alpha\beta(\vec{u} \cdot \vec{v})$

Exercise 2.16. Prove Theorem 2.15.

Next we define **cross product** (**vector product**). Unlike dot product, cross product is defined only in $\mathbb{R}^3$. Moreover cross product of two vectors in a vector.

Definition 2.17. The **cross product** (**vector product**) of two vectors $\vec{v}$ and $\vec{w}$ in $\mathbb{R}^3$, denoted by $\vec{v} \times \vec{w}$, is defined as follows.

$$\vec{v} \times \vec{w} = |\vec{v}||\vec{w}| \sin\theta\vec{n}$$

where θ is the angle between $\vec{v}$ and $\vec{w}$, and $\vec{n}$ is a unit vector perpendicular to $\vec{v}$ and $\vec{w}$, with a direction given by the "right hand rule"

See a visualization of cross product here. Below is a simple formula to compute the vector product $\vec{v} \times \vec{w}$.

$$\vec{v} \times \vec{w} = \begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix} \times \begin{bmatrix} w_1 \\ w_2 \\ w_3 \end{bmatrix} = \begin{bmatrix} v_2w_3 - v_3w_2 \\ v_3w_1 - v_1w3 \\ v_1w_2 - v_2w_1 \end{bmatrix}$$

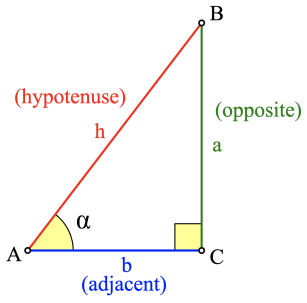
Theorem 2.18. *The vector product satisfies the following properties, where r is a scalar and $\vec{v}$, $\vec{u}$, $\vec{w}$ are vectors in $\mathbb{R}^3$.*

- $\vec{v} \times \vec{u} = -\vec{u} \times \vec{v}$
- $\vec{v} \times \vec{v} = 0$
- $\vec{v} \times (\vec{u} + \vec{w}) = (\vec{v} \times \vec{u}) + (\vec{v} \times \vec{w})$
- $\vec{v} \times r\vec{u} = r(\vec{v} \times \vec{u})$

3 Angles, Projections and Rotations

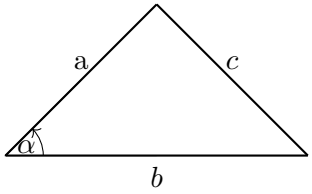
Let us recall the definition of sine and cosine of an angle from the right triangle given below.

$$\sin\alpha = \frac{a}{h}$$
$$\cos\alpha = \frac{b}{h}$$



Next we find an expression for dot product interms of the length of the vectors and the angle between them. Towards that, let us first recall the law of cosines. Consider the triangle given below with lengths of the sides a, b, c. Then,

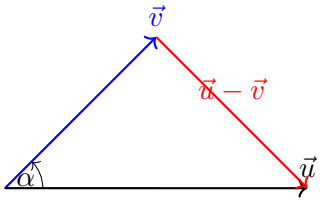
$$c^2 = a^2 + b^2 - 2ab\cos\alpha.$$



Theorem 3.1. *Let $\vec{v}$ and $\vec{u}$ be two vectors in plane. Then $\vec{v}\cdot\vec{u} = |\vec{v}||\vec{u}|\cos\alpha$, where α is the angle between $\vec{v}$ and $\vec{u}$.*

Proof. Let $\vec{u} = \begin{bmatrix} u_1 \\ u_2 \end{bmatrix}$ and $\vec{v} = \begin{bmatrix} v_1 \\ v_2 \end{bmatrix}$

Now we draw $\vec{u}$, $\vec{v}$ and $\vec{u} - \vec{v}$.



By low of cosines we get

$$|\vec{u} - \vec{v}|^2 = |\vec{u}|^2 + |\vec{v}|^2 - 2|\vec{u}||\vec{v}|\cos\alpha \tag{3.1}$$

The left side of (3.1) is equal to

$$(u_1 - v_1)^2 + (u_2 - v_2)^2 = u_1^2 + u_2^2 + v_1^2 + v_2^2 - 2u_1v_1 - 2u_2v_2.$$

The right side of (3.1) is equal to

$$u_1^2 + u_2^2 + v_1^2 + v_2^2 - 2|\vec{u}||\vec{v}|\cos\alpha$$

By equating both the expressions, we get,

$$\begin{aligned} |\vec{u}||\vec{v}|\cos\alpha &= u_1v_1 + u_2v_2 \\ &= \vec{v}\cdot\vec{u} \end{aligned}$$

□

3.1 Linear Transformations

In linear algebra we analyze certain functions where the domain and range are $\mathbb{R}^2$ or $\mathbb{R}^3$.

Definition 3.2. A function f with domain and codomain $\mathbb{R}^2$ or $\mathbb{R}^3$ is called **linear transformation** if the following holds. For any two vectors $\vec{x}$ and $\vec{y}$, and a scaler r,

- $f(\vec{x}+\vec{y}) = f(\vec{x}) + f(\vec{y})$
- $f(r\vec{x}) = rf(\vec{x})$

An important property of a linear transformation f is that we can associate it with a matrix M_f such that $f(\vec{x}) = M_f\vec{x}$. Remember that $\vec{x}$ is a column vector here. We say that M_f represents the linear transformation f. Let us prove this fact for linear transformations from $\mathbb{R}^2$ to $\mathbb{R}^2$.

Theorem 3.3. *For any linear transformation f from $\mathbb{R}^2$ to $\mathbb{R}^2$, there is a matrix M_f such that $f(\vec{x}) = M_f\vec{x}$.*

Proof. Let $\vec{e}_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$ and $\vec{e}_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$. The set $\{\vec{e}, \vec{e_2}\}$ is called *standard basis* for $\mathbb{R}^2$. Any vector $\vec{v} = \begin{bmatrix} a_1 \\ a_2 \end{bmatrix}$ can be written as *linear combination* of $\vec{e}_1$ and $\vec{e}_2$. That is,

$$\vec{v} = a_1\vec{e}_1 + a_2\vec{e}_2.$$

Let $f(\vec{e_1}) = \begin{bmatrix} m_{1,1} \\ m_{2,1} \end{bmatrix}$ and $f(\vec{e_2}) = \begin{bmatrix} m_{1,2} \\ m_{2,2} \end{bmatrix}$. Now we claim that the following matrix represents the linear transformation f.

$$M_f = \begin{bmatrix} m_{1,1} & m_{1,2} \\ m_{2,1} & m_{2,2} \end{bmatrix}.$$

Fix an arbitrary vector $\vec{x} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$. Then,

$$\begin{aligned} f(\vec{x}) &= f\left(x_1 \begin{bmatrix} 1 \\ 0 \end{bmatrix} + x_2 \begin{bmatrix} 0 \\ 1 \end{bmatrix}\right) \\ &= f\left(x_1 \begin{bmatrix} 1 \\ 0 \end{bmatrix}\right) + f\left(x_2 \begin{bmatrix} 0 \\ 1 \end{bmatrix}\right) \\ &= x_1 f\left(\begin{bmatrix} 1 \\ 0 \end{bmatrix}\right) + x_2 f\left(\begin{bmatrix} 0 \\ 1 \end{bmatrix}\right) \end{aligned}$$

$$
\begin{aligned}
&= x_1 \begin{bmatrix} m_{1,1} \\ m_{2,1} \end{bmatrix} + x_2 \begin{bmatrix} m_{1,2} \\ m_{2,2} \end{bmatrix} \\
&= \begin{bmatrix} m_{1,1} \cdot x_1 \\ m_{2,1} \cdot x_2 \end{bmatrix} + \begin{bmatrix} m_{1,2} \cdot x_2 \\ m_{2,2} \cdot x_2 \end{bmatrix} \\
&= \begin{bmatrix} m_{1,1} \cdot x_1 + m_{1,2} \cdot x_2 \\ m_{2,1} \cdot x_2 + m_{2,2} \cdot x_2 \end{bmatrix} \\
&= \begin{bmatrix} m_{1,1} & m_{1,2} \\ m_{2,1} & m_{2,2} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \\
&= M_f \vec{x}.
\end{aligned}
$$

This completes the proof. □

For the examples of linear transformations we discuss here, we also come up a matrix that represents the linear transformation.

Example 3.4. We have seen scalar multiplication of a vector. Let r be a scalar and $\vec{u} = \begin{bmatrix} u_1 \\ u_2 \\ u_3 \end{bmatrix}$ be a vector. Then the scalar multiplication of r and $\vec{u}$ is

$$r\vec{u} = \begin{bmatrix} ru_1 \\ ru_2 \\ ru_3 \end{bmatrix}$$

This can be thought of as function that scales each vector by a factor of r. This can be represented by the following matrix.

$$\begin{bmatrix} r & 0 & 0 \\ 0 & r & 0 \\ 0 & 0 & r \end{bmatrix}$$

Notice that

$$\begin{bmatrix} r & 0 & 0 \\ 0 & r & 0 \\ 0 & 0 & r \end{bmatrix} \begin{bmatrix} u_1 \\ u_2 \\ u_3 \end{bmatrix} = \begin{bmatrix} ru_1 \\ ru_2 \\ ru_3 \end{bmatrix}$$

In the case of $\mathbb{R}^2$, the matrix is a diagonal matrix of order 2×2 and the diagonal entries are equal.

3.1.1 Scaling Transformation

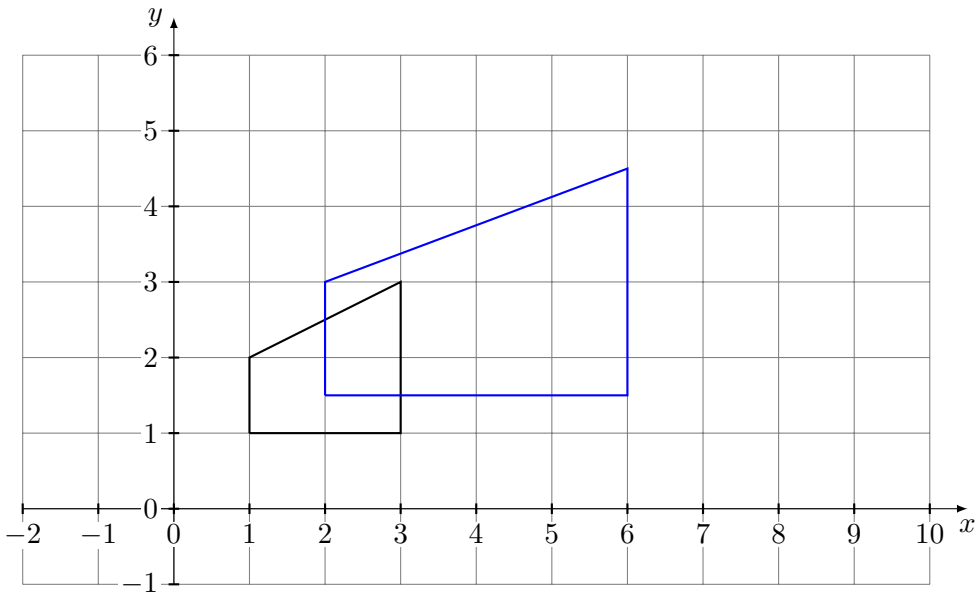
In Example 3.4 we scales the vector by a factor of r on all the components. A transformation, called **scaling transformation**, where we could scale the vector differently in different components. A scaling transformation is represented by the following matrix.

$$\begin{bmatrix} r_1 & 0 & 0 \\ 0 & r_2 & 0 \\ 0 & 0 & r_3 \end{bmatrix}$$

In the case of $\mathbb{R}^2$, the matrix is a diagonal matrix of order 2×2.

Example 3.5. Below is an example of scaling transformation with the scaling matrix

$$\begin{bmatrix} 2 & 0 \\ 0 & 1.5 \end{bmatrix}$$



3.1.2 Orthogonal Projection.

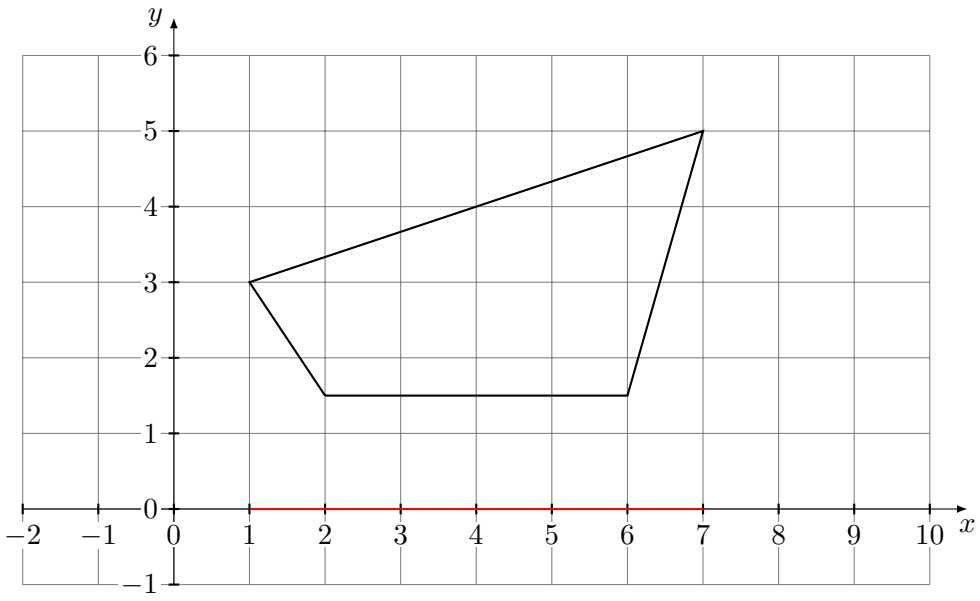
A function that maps a vector $\vec{u} = \begin{bmatrix} u_1 \\ u_2 \\ u_3 \end{bmatrix}$

to $\begin{bmatrix} u_1 \\ u_2 \\ 0 \end{bmatrix}$ is an orthogonal projection to xy-plane. This function is represented by the matrix

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

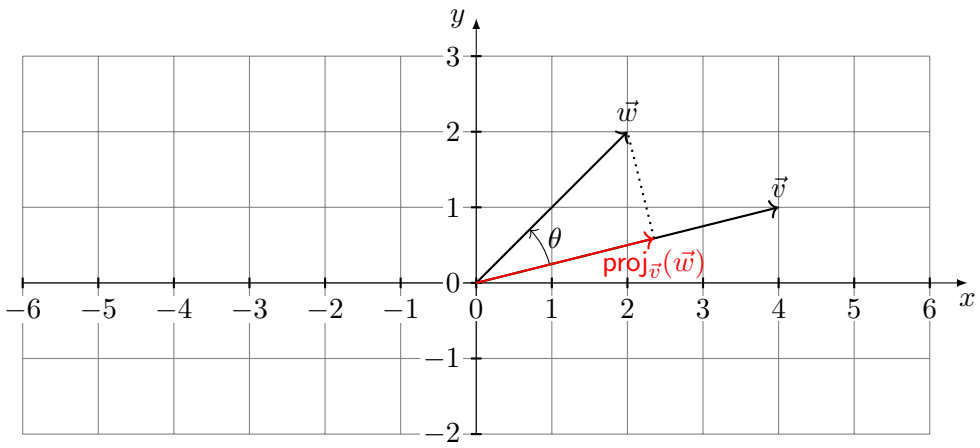
Example 3.6. Below is an example of orthogonal projection to the x-axis in $\mathbb{R}^2$. The matrix associated with it is

$$\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$$



3.1.3 Projection to a vector

Projection of a vector $\vec{w}$ on a vector $\vec{v}$ is a shadow which $\vec{w}$ casts on $\vec{v}$ due to a light ray perpendicular to $\vec{v}$ that hits $\vec{w}$. We denote the projection of a vector $\vec{w}$ on a vector $\vec{v}$ by $\mathsf{proj}_{[\vec{v}]}(\vec{w})$. In the below figure it is the vector in the red color. This is a vector in the direction of $\vec{v}$ and its length is $|\vec{w}| \cos\theta$.

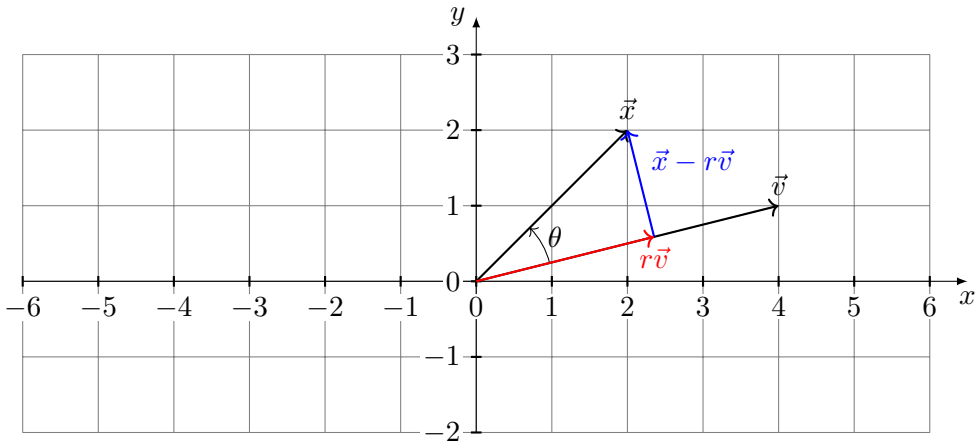


Let $\vec{v}$ be a fixed vector. Then, $\mathsf{proj}_{[\vec{v}]}$ is a function from $\mathbb{R}^2$ to $\mathbb{R}^2$.

Theorem 3.7. *The projection of $\vec{x}$ on to a nonzero vector $\vec{v}$, is*

$$\mathsf{proj}_{[\vec{v}]}(\vec{x}) = \frac{\vec{v} \cdot \vec{x}}{\vec{v} \cdot \vec{v}} \vec{v}$$

Proof. Let $\vec{x}$ be a vector in $\mathbb{R}^2$. We know that $\mathsf{proj}_{[\vec{v}]}(\vec{x})$ is a vector in the direction $\vec{v}$. That is, $\mathsf{proj}_{[\vec{v}]}(\vec{x}) = r\vec{v}$ for some scalar r. Now consider the vector $\vec{x} - r\vec{v}$ as shown in the figure below.



The vector $\vec{x} - r\vec{v}$ is perpendicular to $\vec{v}$. That is, the angle between $\vec{x} - r\vec{v}$ and $\vec{v}$ is $\pi/2$. Thus, by Theorem 3.1, we know that their dot product is 0. That is,

$$\begin{aligned} \vec{v} \cdot (\vec{x} - r\vec{v}) &= 0 \\ \vec{v} \cdot \vec{x} - r(\vec{v} \cdot \vec{v}) &= 0 \qquad \text{(By Theorem 2.15)} \\ r &= \frac{\vec{v} \cdot \vec{x}}{\vec{v} \cdot \vec{v}} \end{aligned}$$

This implies that

$$\mathsf{proj}_{[\vec{v}]}(\vec{x}) = \frac{\vec{v} \cdot \vec{x}}{\vec{v} \cdot \vec{v}} \vec{v}$$

□

The concept of projection extend to higher dimension.

Example 3.8. In $\mathbb{R}^3$, the orthogonal projection of a vector

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix}$$

into y-axis is

$$\frac{\begin{bmatrix} x \\ y \\ z \end{bmatrix} \cdot \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}}{\begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \cdot \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}} \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} = \frac{y}{1} \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ y \\ 0 \end{bmatrix}$$

This matches with our intuitive expectation.

Next we show that projection is a linear transformation.

Theorem 3.9. *Let $\vec{v}$ be a fixed vector in $\mathbb{R}^2$. Then, $\mathsf{proj}_{\vec{v}}$ is a linear transformation.*

Proof. Let $\vec{x}$ and $\vec{y}$ be two vectors in $\mathbb{R}^2$. Then,

$$\begin{aligned} \mathsf{proj}_{\vec{v}}(\vec{x} + \vec{y}) &= \frac{\vec{v} \cdot (\vec{x} + \vec{y})}{\vec{v} \cdot \vec{v}} \vec{v} \\ &= \frac{(\vec{v} \cdot \vec{x}) + (\vec{v} \cdot \vec{y})}{\vec{v} \cdot \vec{v}} \vec{v} \qquad \text{(By Theorem 2.15)} \\ &= \frac{(\vec{v} \cdot \vec{x})}{\vec{v} \cdot \vec{v}} \vec{v} + \frac{(\vec{v} \cdot \vec{y})}{\vec{v} \cdot \vec{v}} \vec{v} \qquad \text{(By Lemma 2.10)} \\ &= \mathsf{proj}_{\vec{v}}(\vec{x}) + \mathsf{proj}_{\vec{v}}(\vec{y}) \end{aligned}$$

Let r be a scalar and $\vec{x}$ be a vector. Then

$$\begin{aligned}\mathsf{proj}_{\vec{v}}(r\vec{x}) &= \frac{\vec{v}\cdot r\vec{x}}{\vec{v}\cdot\vec{v}}\vec{v} \\ &= \frac{r(\vec{v}\cdot\vec{x})}{\vec{v}\cdot\vec{v}}\vec{v} \\ &= r\mathsf{proj}_{\vec{v}}(\vec{x})\end{aligned}$$

□

Next, we construct a matrix $P_{\vec{v}}$ such that $\mathsf{proj}_{\vec{v}}(\vec{x}) = P_{\vec{v}}\vec{x}$. Let

$$\vec{v} = \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} \qquad \text{and} \qquad \vec{x} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}.$$

Then,

$$\begin{aligned}\mathsf{proj}_{[\vec{v}]}(\vec{x}) &= \frac{\vec{v}\cdot\vec{x}}{\vec{v}\cdot\vec{v}}\vec{v} \\ &= \frac{\begin{bmatrix} v_1 \\ v_2 \end{bmatrix}\cdot\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}}{\begin{bmatrix} v_1 \\ v_2 \end{bmatrix}\cdot\begin{bmatrix} v_1 \\ v_2 \end{bmatrix}}\begin{bmatrix} v_1 \\ v_2 \end{bmatrix} \\ &= \frac{v_1x_1 + v_2x_2}{v_1^2 + v_2^2}\begin{bmatrix} v_1 \\ v_2 \end{bmatrix} \\ &= \left(\frac{v_1}{|\vec{v}|^2}x_1 + \frac{v_2}{|\vec{v}|^2}x_2\right)\begin{bmatrix} v_1 \\ v_2 \end{bmatrix} \\ &= \begin{bmatrix} \frac{v_1^2}{|\vec{v}|^2}x_1 + \frac{v_1v_2}{|\vec{v}|^2}x_2 \\ \frac{v_1v_2}{|\vec{v}|^2}x_1 + \frac{v_2^2}{|\vec{v}|^2}x_2 \end{bmatrix} \\ &= \begin{bmatrix} \frac{v_1^2}{|\vec{v}|^2} & \frac{v_1v_2}{|\vec{v}|^2} \\ \frac{v_1v_2}{|\vec{v}|^2} & \frac{v_2^2}{|\vec{v}|^2} \end{bmatrix}\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}\end{aligned}$$

So, we got

$$P_{\vec{v}} = \begin{bmatrix} \frac{v_1^2}{|\vec{v}|^2} & \frac{v_1v_2}{|\vec{v}|^2} \\ \frac{v_1v_2}{|\vec{v}|^2} & \frac{v_2^2}{|\vec{v}|^2} \end{bmatrix}.$$

The matrix $P_{\vec{v}}$ is called the projection matrix. Another useful way to write the projection matrix $P_{\vec{v}}$ is as follows.

$$P_{\vec{v}} = \begin{bmatrix} \frac{v_1^2}{|\vec{v}|^2} & \frac{v_1 v_2}{|\vec{v}|^2} \\ \frac{v_1 v_2}{|\vec{v}|^2} & \frac{v_2^2}{|\vec{v}|^2} \end{bmatrix} = \frac{1}{|\vec{v}|^2}\left(\begin{bmatrix} v_1 \\ v_2 \end{bmatrix} \begin{bmatrix} v_1 & v_2 \end{bmatrix} \right) = \frac{1}{|\vec{v}|^2}(\vec{v}\vec{v}^T).$$

For a vector $u \in R^3$, the projection matrix $P_{\vec{u}}$ is defined as follows.

$$P_{\vec{u}} = \frac{1}{|\vec{u}|^2}(\vec{u}\vec{u}^T).$$

The projection matrix has an important property. The square of a projection matrix is the projection matrix itself.

Theorem 3.10. *Let $\vec{u}$ be a vector in $\mathbb{R}^3$. Then $P_{\vec{u}}^2 = P_{\vec{u}}$.*

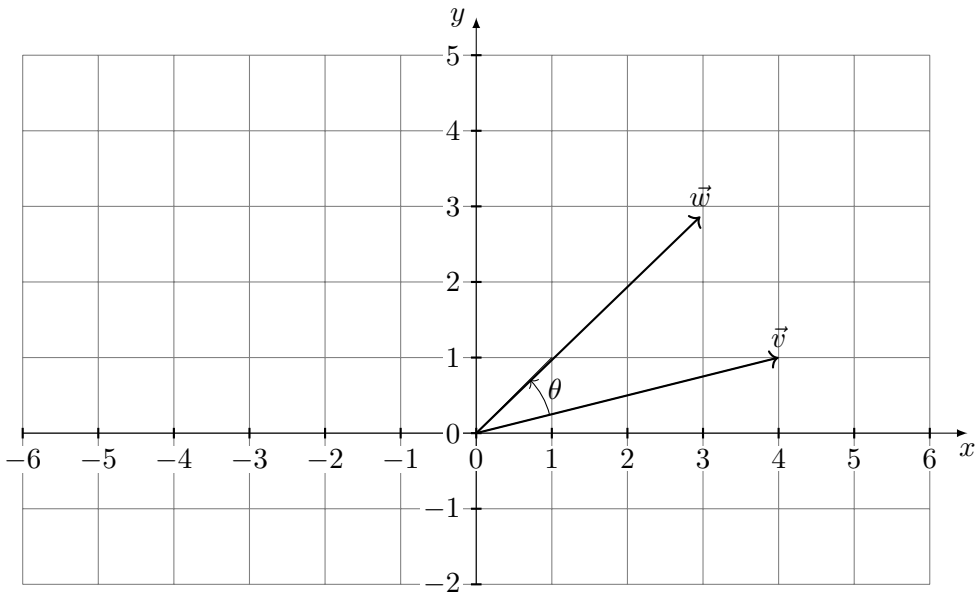
Proof. Let $\vec{u}^T = [u_1 \quad u_2 \quad u_3]$.

$$\begin{aligned}
P_{\vec{u}}^2 &= \frac{1}{|\vec{u}|^2}(\vec{u}\vec{u}^T)\frac{1}{|\vec{u}|^2}(\vec{u}\vec{u}^T) \\
&= \frac{1}{|\vec{u}|^4}\vec{u}(\vec{u}^T\vec{u})\vec{u}^T \\
&= \frac{1}{|\vec{u}|^4}\begin{bmatrix} u_1 \\ u_2 \\ u_3 \end{bmatrix}\left(\begin{bmatrix} u_1 & u_2 & u_3 \end{bmatrix} \begin{bmatrix} u_1 \\ u_2 \\ u_3 \end{bmatrix} \right) \begin{bmatrix} u_1 & u_2 & u_3 \end{bmatrix} \\
&= \frac{1}{|\vec{u}|^4}\begin{bmatrix} u_1 \\ u_2 \\ u_3 \end{bmatrix}[|\vec{u}|^2]\begin{bmatrix} u_1 & u_2 & u_3 \end{bmatrix} \\
&= \frac{1}{|\vec{u}|^4}\begin{bmatrix} |\vec{u}|^2 u_1 \\ |\vec{u}|^2 u_2 \\ |\vec{u}|^2 u_3 \end{bmatrix}\begin{bmatrix} u_1 & u_2 & u_3 \end{bmatrix} \\
&= \frac{|\vec{u}|^2}{|\vec{u}|^4}\begin{bmatrix} u_1 \\ u_2 \\ u_3 \end{bmatrix}\begin{bmatrix} u_1 & u_2 & u_3 \end{bmatrix} \\
&= \frac{1}{|\vec{u}|^2}(\vec{u}\vec{u}^T) \\
&= P_{\vec{u}}
\end{aligned}$$

This completes the proof. □

3.1.4 Rotation

Consider $\mathbb{R}^2$. Let $\vec{v}$ be a vector. Then the rotation of $\vec{v}$ in the anticlockwise direction moves the vector as shown in the figure.



This is a linear transformation. We find a matrix that represent the rotation by θ radians anticlockwise wise. Let $\vec{v} = \begin{bmatrix} v_1 \\ v_2 \end{bmatrix}$. Let $\vec{w} = \begin{bmatrix} w_1 \\ w_2 \end{bmatrix}$ be the vector we get after rotation. Let $r = |\vec{v}| = |\vec{w}|$ and α be the angle between x-axis and $\vec{v}$.

Then,

$$\begin{aligned} v_1 &= r\cos\alpha \\ v_2 &= r\sin\alpha \\ w_1 &= r\cos(\alpha+\theta) \\ w_2 &= r\sin(\alpha+\theta) \end{aligned}$$

Now we want to write w_1 and w_2 in terms of v_1 and v_2.

$$\begin{aligned} w_1 &= r\cos(\alpha+\theta) \\ &= r(\cos\alpha\cos\theta - \sin\alpha\sin\theta) \\ &= v_1\cos\theta - v_2\sin\theta \\ w_2 &= r(\sin\alpha\cos\theta + \cos\alpha\sin\theta) \\ &= v_1\sin\theta + v_2\cos\theta \end{aligned}$$

Thus, we can write

$$\begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} w_1 \\ w_2 \end{bmatrix}$$

So, the function that rotates vectors θ radian in the anticlockwise direction is represented by the matrix

$$\begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}$$

Example 3.11. Below is an example rotation by $\pi/6$ radians in the anticlockwise direction in $\mathbb{R}^2$. The matrix that represents this rotation is

$$\begin{bmatrix} \cos\pi/6 & -\sin\pi/6 \\ \sin\pi/6 & \cos\pi/6 \end{bmatrix}$$

